# PR Opportunity Monitor

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# The Problem

A small Arizona PR agency manages media relations for attorneys, healthcare providers, and telecom companies.

**They only catch stories they already thought to search for**

- Adjacent opportunities are missed entirely

**They surface stories where the client already appears**

- Competitor coverage, not new opportunities

# The Insight

*The Difference Between Reactive and Proactive*

| Keyword Alerts | Semantic Matching |
|---|---|
| Finds stories that mention your topic | Finds stories your client *could enter* |
| Surfaces existing coverage | Surfaces missing voices |
| You define the vocabulary | The meaning defines the match |

Keyword alerts tell you what's already been said. Semantic matching tells you where your client could speak next.

# The Four Clients & Their Data

| Client | Industry | Corpus | Notable |
|---|---|---|---|
| Client 1 | Personal Injury Law | 125 articles, 28 topics | Highest quality: written drafts provided directly by client |
| Client 2 | Healthcare | 94 + 47 articles | Smallest real corpus: supplemented with health awareness calendar |
| Client 3 | Telecom | 613 articles, 29 topics | Largest corpus: scraped from real placements |
| Client 4 | Business Law | 194 articles | Heavy commercial real estate & employment law coverage |

*Articles span 2020–2025, collected through web scraping, browser automation, and direct client submissions.*

# Corpus Construction

*How We Built the Expertise Fingerprints*

***Three-stage pipeline:*** newspaper3k → Playwright fallback → direct client drafts (Client 1 only)

154/901 articles regex-cleaned, 100-word minimum filtered paywalled stubs

****Client 2 special case:*** 94 articles wasn't enough. Wrote 47 seed documents covering their 2026 health awareness calendar to fill topic gaps.

| Client | Articles | Source |
|---|---|---|
| Client 1 | 125 | Client-provided drafts |
| Client 2 | 94 + 47 seeds | Scraped + written |
| Client 3 | 613 | Scraped |
| Client 4 | 194 | Scraped |

# The System

01

### Client Article Corpus

94–613 real media placements per client spanning 3–5 years of coverage

CORPUS

→

02

### Bi-Encoder Embeddings

384-dim dense vectors via all-MiniLM-L6-v2 — normalized for cosine similarity

SEMANTIC

→

03

### ChromaDB Vector Store

One isolated collection per client — expertise fingerprint prevents cross-client contamination

ISOLATED

→

04

### Incoming News

15–45 weekly headlines via Google News RSS — press releases filtered out

NO RATE LIMITS

→

05

### Semantic Match

Dense-only retrieval (α = 1.0) — outperforms hybrid search across all clients

DENSE RETRIEVAL

→

06

### RAG Pitch Draft

Grounded in K = 3 retrieved articles — ready for practitioner review & send

LLAMA 3.3 70B

α = 1.0 (dense only) · K = 3 retrieved articles · Strong ≥ 0.55 · Adjacent 0.40–0.55 · Up to 5 strong matches surfaced per client per run

# Hybrid vs. Dense Retrieval

*The Key Technical Decision*

We tested hybrid retrieval (dense + TF-IDF) across a grid of alpha values for all 4 clients:

| α | Retrieval Mix |
|---|---|
| 0.3 | Mostly sparse (TF-IDF) |
| 0.5 | Equal blend |
| 0.9 | Mostly dense |
| **1.0** | **Dense only ← winner** |

***Assumption:*** hybrid would outperform dense, especially for legal and medical terminology.

***Result:*** α = 1.0 won for all 4 clients.

***Why:*** journalistic language varies too much for keyword overlap to add signal.

# Evaluation

*Leave-One-Out Retrieval Consistency*

For each article: use its title as a query, exclude the article itself, count a hit if any result scores ≥ 0.55

Leave-one-out because the bi-encoder is frozen; evaluating the index, not training a model

| Client | Consistency (α = 1.0) |
|---|---|
| Client 1 | 0.875 |
| Client 2 | 0.847 |
| Client 3 | 0.798 |
| Client 4 | 0.617 |

# Demo

# Key Findings

- Dense retrieval outperformed hybrid for all 4 clients ($\alpha$=1.0)
- Retrieval consistency scores at $\alpha$ = 1.0: Client 1 0.875, Client 2 0.847, Client 3 0.798, Client 4 0.617
- Client 2 calendar seeds successfully extended coverage to topics with no prior media presence
- Threshold tuning (0.50 → 0.55) meaningfully reduced false positives (Arizona sports stories)
- System surfaces ~3-10 actionable pitches per client per week

# Future Improvements

- **Automate the weekly pipeline** - currently requires manual notebook execution; schedule to run and push to the hosted UI without technical setup
- **DPO reranking** - feedback infrastructure is live but not yet used; accumulated preference pairs could fine-tune a client-specific re-ranker
- **Corpus refresh** - static index requires manual re-indexing; monthly or quarterly updates needed to keep expertise fingerprints current

# Thank you!